

Topic

SGLang Project and Roadmap

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SGLang Developer





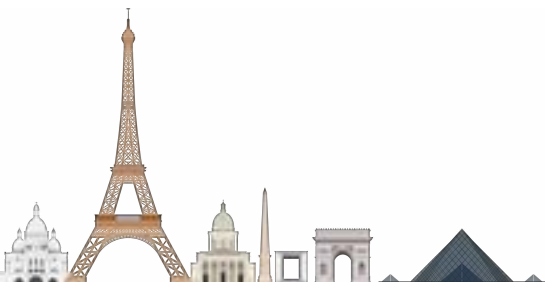
SGLang Roadmap

Xinyuan Tong
May 5th, 2026

- 01 **SGLang Overview**

- 02 **Recent Feature Highlight**

- 03 **Future Roadmap (2026 Q2)**



SGLang: Overview

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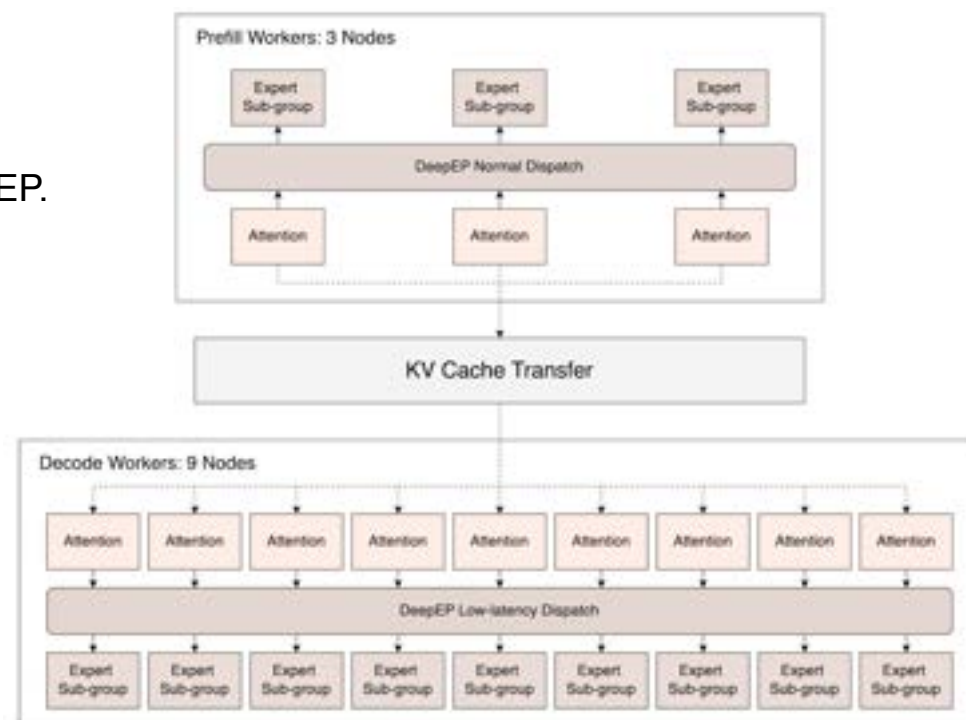
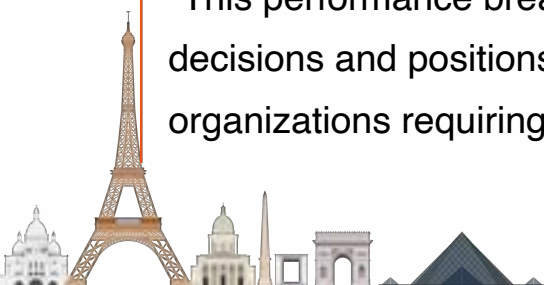
Industry-First Performance

SGLang is the first open-source system that nearly match the performance of DeepSeek official blog with PD disaggregation and EP.

Performance Metrics (May 2025)

- ❑ 52.3k input tokens/s/node industry-leading input processing speed
- ❑ 22.3k output tokens/s/node Exceptional generation throughput
- ❑ 5x cost reduction vs. DeepSeek API pricing
- ❑ 10+ teams successfully reproduced results

"This performance breakthrough validates our architectural decisions and positions SGLang as the go-to solution for organizations requiring enterprise-scale AI inference capabilities."



Multiple Hardware Support

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H20

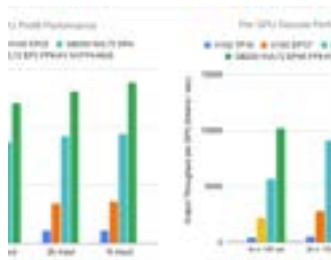


Best Practices for Serving DeepSeek R1 on H20

Together with SGLang: Best Practices for Serving DeepSeek-R1 on H20-96G

by: Tianyu Zhang*, Peng Zhang*, Xuesong Gao, Yun Zhang, September 26, 2025

Introduction Operationalizing scaled Mixture-of-Experts (MoE) models such as DeepSeek-R1 requires a careful balance of latency, throughput, and cost. The challenge is especially acute on hardware with asymmetric performance profiles—for example, the H20 GPU, which offers high memory bandwidth but co...



Deploying DeepSeek on GB200 NVL72 with PD and Large Scale EP (Part II): 3.8x Prefill, 4.8x Decode Throughput

by: The SGLang Team, September 25, 2025

The GB200 NVL72 is one of the most powerful hardware for deep learning. In this blog post, we share our progress after our previous blog post to optimize the inference performance of DeepSeek V3/R1 with FP8 attention, NVFP4 MoE, large-scale expert parallelism, prefill-decode disaggregation, and vari...

NVL72



Optimizing FP4 Mixed-Precision Inference on AMD GPUs

by: Haoxul Mai, Lei Zhang, September 21, 2025

Introduction As frontier large language models (LLMs) continue scaling to unprecedented sizes, they demand increasingly more compute power and memory bandwidth from GPUs. Both GPU manufacturers and model developers are shifting toward low-precision floating-point formats. FP4 (4-bit floating point) ...

JAX

SGL-JAX: High-Performance LLM Inference on JAX/TPU

SGL-JAX is a high-performance, JAX-based inference engine for Large Language Models (LLMs), specifically optimized for Google TPUs. It is engineered from the ground up to deliver exceptional throughput and low latency for the most demanding LLM serving workloads.

The engine incorporates state-of-the-art techniques to maximize hardware utilization and serving efficiency, making it ideal for deploying large-scale models in production on TPUs.



Spark



NVIDIA DGX Spark In-Depth Review: A New Standard for Local AI Inference

by: Jerry Zhou and Richard Chen, October 13, 2025

Thanks to NVIDIA's early access program, we are thrilled to get our hands on the NVIDIA DGX™ Spark. It's quite an unconventional system, as NVIDIA rarely releases compact, all-in-one machines that bring supercomputing-class performance to a desktop workstation form factor. Over the past year, SGLang...

Intel



Cost Effective Deployment of DeepSeek R1 with Intel® Xeon® 6 CPU on SGLang

by: Intel PyTorch Team, July 14, 2025

The impressive performance of DeepSeek R1 marked a rise of giant Mixture of Experts (MoE) models in Large Language Models (LLM). However, its massive model size and unique architecture have posed new challenges on deployment. The significant memory requirements will normally require 8x or even 16x h...

SemiAnalysis



SGLang and NVIDIA Accelerating SemiAnalysis InferenceMAX and GB200 Together

by: NVIDIA and community SGLang developers, Oct 14, 2025

The SGLang and NVIDIA teams have a strong track record of collaboration, consistently delivering inference optimizations and system-level improvements to ensure exceptional performance of the SGLang framework. Most recently, this collaboration has been centered on the NVIDIA Blackwell architecture, ...

Community Growth & Industry Adoption



Developer Community Expansion

1000+ Contributors

Active developers contributing code, documentation, and community support

60+
Institution

Universities, research labs, and companies actively using SGLang

20+
Enterprise Users

Companies adopting SGLang as their default DeepSeek inference engine in the first month of release

Recent Feature Highlight

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Zero-overhead Speculative Decoding

Tune the scheduler for speculative decoding and seeing 10% - 20% speedup across the board.

SGL Diffusion

Accelerate image and video generation for diffusion models for production-level serving

Spec Forge v0.2

Draft Model Training Framework

Mini-SGLang

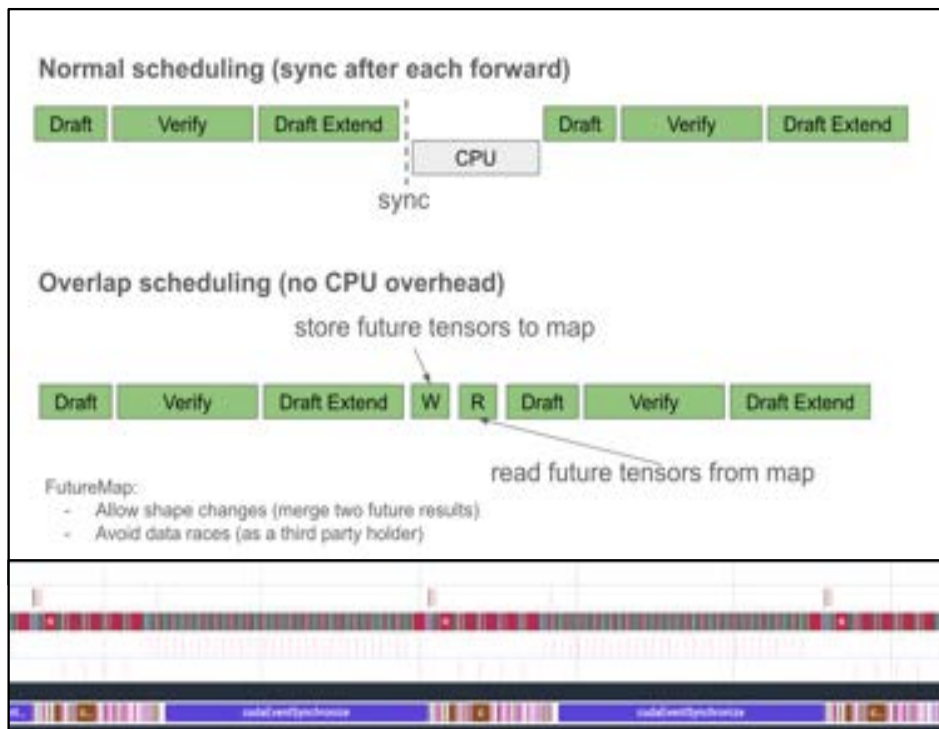
A lightweight yet high-performance inference framework sharing the high-level system architectures as SGLang

EPD Disaggregation

Optimized data handling for large-scale multimodal models deployment.



Zero-overhead Speculative Decoding



Zero-overhead CPU runtime for LLM

SGLang has been **pioneering** the zero-overhead CPU runtime for LLM runtime last year.

Scheduler Tuning for Spec Decoding

~20% speedup across the board.

Stream Design

GPU Forward Stream: Handle all tensor forwarding without GPU blocking

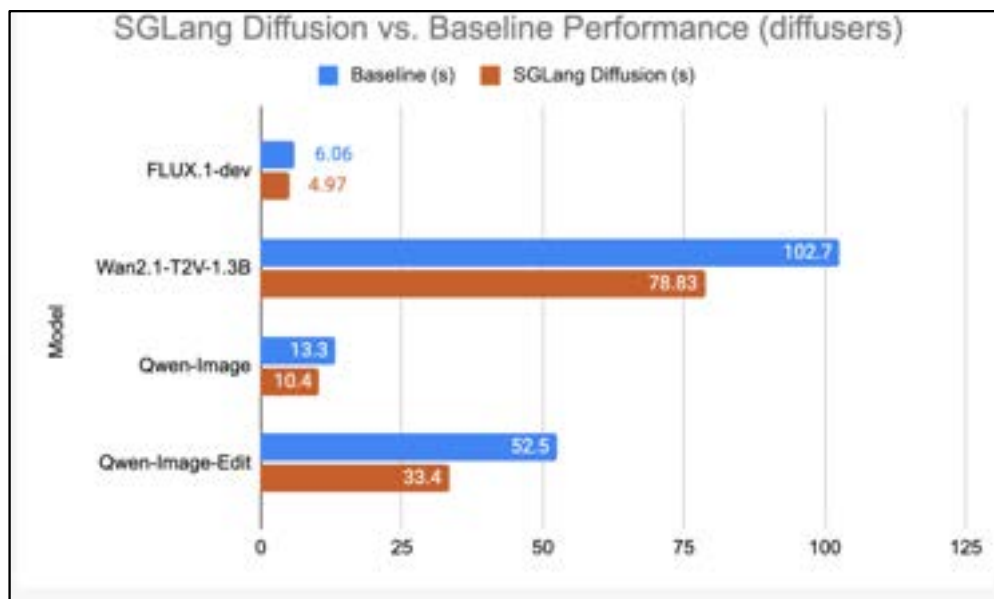
CPU Schedule Stream: Delay one step output process and schedule continuous batching

<https://lmsys.org/blog/2024-12-04-sglang-v0-4/>

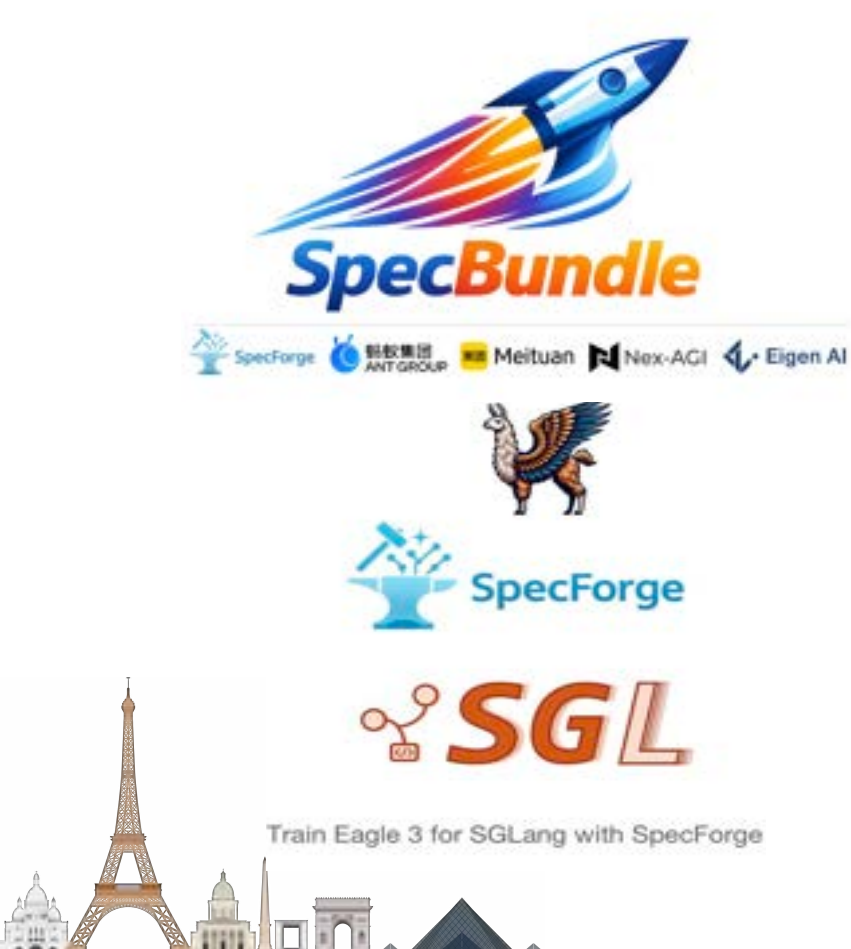


SGLang Diffusion

- **1.2–5.9×** speedup for image/video generation
- Supports autoregressive + diffusion multimodal models (Wan, Hunyuan, Qwen-Image, Flux)
- **Optimized kernels + parallelism** techniques
- Collaboration: **FastVideo team, AntGroup**



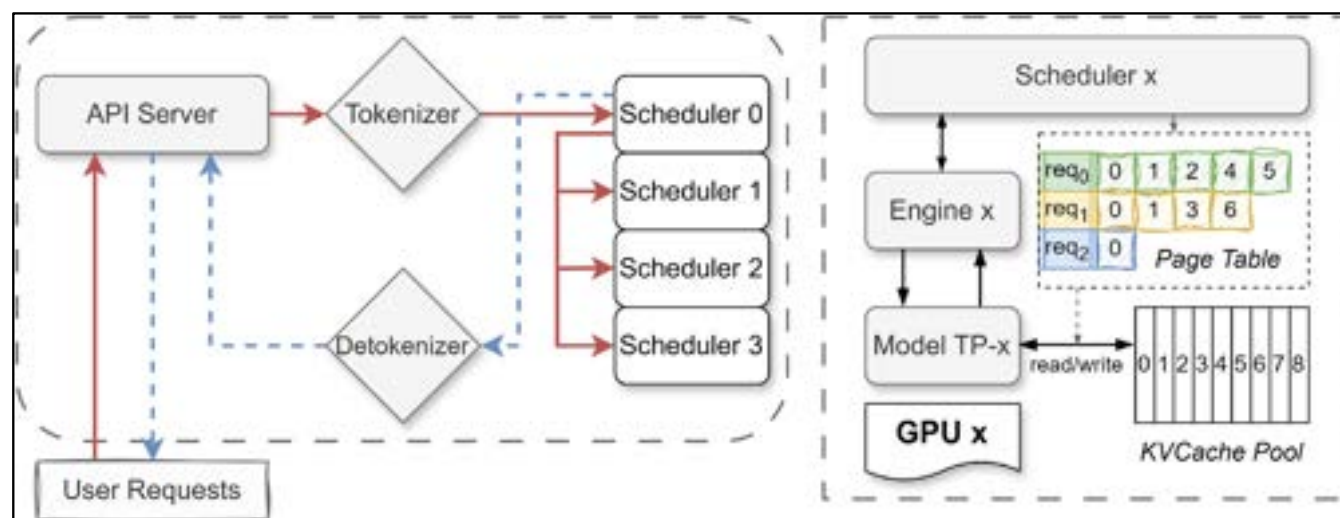
 SGL DIFFUSION



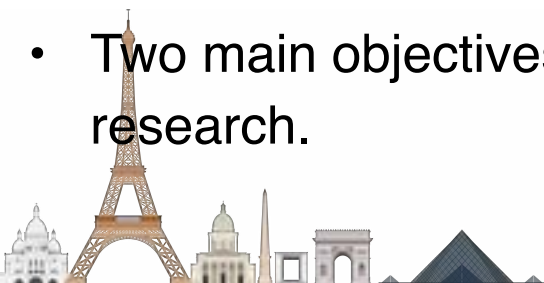
SpecForge: Draft Model Training Framework

Native SGLang integration for speculative decoding optimization

- Training: **Scalable distributed + memory-efficient**
- SpecBundle: Production-grade EAGLE-3 checkpoints on **large-scale** datasets
- Model Support: Llama 4, DeepSeek, Qwen3 MoE, GTP-OSS
- Deployments: Ant, Meituan, Nex-AGI, EigenAI



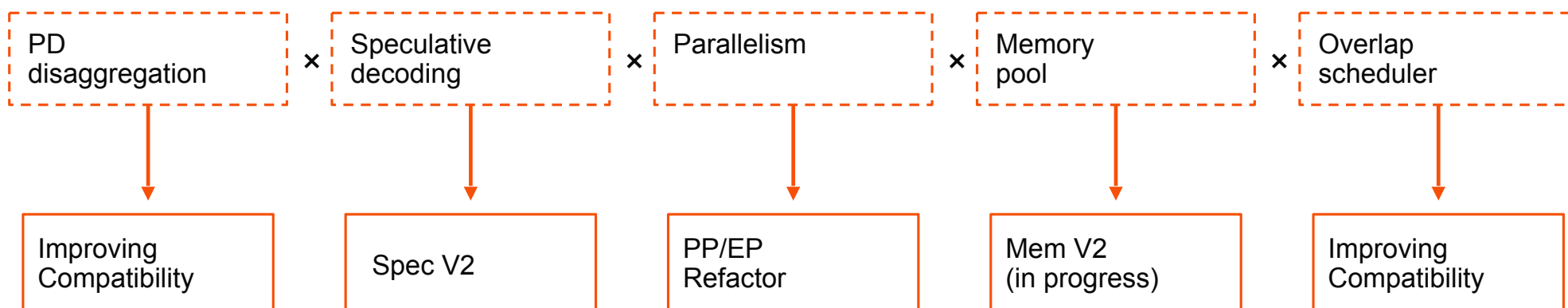
- A **lightweight** yet **high-performance** inference framework sharing the high-level system architectures as SGLang
- Two main objectives: providing **learning** resources and enabling fast **prototyping** for research.



Roadmap (2026 Q2)

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- Feature Compatibility and Reliability: a performant combination of all major features
- Make all advanced features compatible with each other



Usability

Easy install and large-scale deployment across NV/AMD/TPU/CPU

Kernels

Next-gen hardware optimization: GB300/GB200, B300/B200, MI350/MI355, TPU

Multimodal + RL

Diffusion / Omni model support and rollout API integration



Q2 Execution Matrix

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Core refactors

- Scheduler refactor
- KV cache management
- Spec decoding abstraction
- P/D disaggregation
- API server + Rust migration
- CUDA graph runner backend

Parallelism + hardware

- Pipeline / expert / context parallelism
- DP attention + DWDP
- GB200/GB300 NVL72 optimization
- NVIDIA / AMD / TPU / NPU / Intel CPU/XPU

Multimodal + RL

- Diffusion + multimodal generation
- MLLM / VLM perception
- SGLang Omni Stage→Engine refactor
- Miles / slime / verl / AReal rollout API

Reliability + release

- Per-request tracing
- Runtime memory-pool checks
- PD transfer / weight checksum
- CI coverage and reduced flakiness

Source: github.com/sgl-project/sglang/issues/22949 — Development Roadmap (2026 Q2)



Question & Answer

<https://www.sglang.io/>

X: <https://x.com/lmsysorg>



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Panel

Open Discussion & Ecosystem Roundtable
**Open Compute, Open Evaluation, and
Open Governance for Agentic AI**



Serge Stinckwich

Senior Research Fellow, UNU



Dr. Gábor Soós

Advisory Committee Member,
ICCSD UNESCO



Yonghua Lin

VP & Chief Engineer, BAAI

